

Lab 3: Express.js, File System (I/O), and REST API

Overview: Build a RESTful API using Express.js and the File System module with Users and Posts.

Prerequisites

- Completion of Lab 2
- Basic JavaScript knowledge
- Node.js
- Express.js basics
- REST APIs
- JSON files

Learning Objectives

- Build REST APIs
- Use async File System
- Implement CRUD
- Organize Express project
- Understand One-to-Many
- Design reusable schemas
- Prepare for Login/Register

Project Setup

npm init

npm install express

Project Structure (MVC)

controllers/

models/

routes/

middleware/

utils/

index.js

User CRUD

Method	Endpoint	Description
GET	/users	Get all
GET	/users/:id	Get by ID
POST	/users	Create

PUT	/users/:id	Update
DELETE	/users/:id	Delete

Post CRUD

Method	Endpoint	Description
GET	/posts	Get all
GET	/posts/:id	Get by ID
POST	/posts	Create
PUT	/posts/:id	Update
DELETE	/posts/:id	Delete

Relationship

One User has many Posts. Each Post belongs to one User using userId.

Bonus

- Use Joi validation
- Prevent duplicate emails
- Delete posts when deleting user
- Logger middleware
- Helper utilities
- Query parameters